

KENTRO

The Newsletter of the INSTAP Study Center for East Crete

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THE 2007 ARCHEM SEASON

By Andrew Koh

The year 2007 marked a new chapter in the development of Archaeochemistry Research in the Eastern Mediterranean (ARCHEM), a project devoted to the extraction, analysis and contextualization of organic residues from archaeological artifacts. Thanks to the generosity of INSTAP and the Hellenic Studies Endowment Fund of Wayne State University's Department of Classics, Greek and Latin, we were able to expand the ARCHEM team from one to three by bringing in two graduate students from WSU. Both Jennifer Meyer and Thomas Harwood are now pursuing their MA degrees. By helping with the organic residue extractions and meticulous cataloguing work, both have made themselves invaluable members of the ARCHEM team; furthermore, their involvement has brought about the significant expansion and improvement of the overall project. Later, Julie Bergmann, from the University of California-Davis, also joined the team. She arrived in time to learn the basic methodology of the project and to assist with the Papadiokambos excavations.

The ARCHEM team arrived in Crete on the evening of July 3 and started work the following day at the Coulson Laboratory of INSTAP Study Center

for East Crete. After analyzing data from the previous year, we learned that our previous incorporation of the Büchi polyvap into the conservation process was an unqualified success. This approach resulted in our seamless extraction of 895 samples.

From July 4 to July 25, the ARCHEM team contin-

ued to refine the extraction methodology and to address questions and concerns expressed over the past year. One decision we made at the onset of the summer was to save fil-trands from our analysis of each sample. While the small size of our team in past years had made this method impractical, this year's expansion of the ARCHEM



Thomas Harwood, Jennifer Meyer, and Andrew Koh

project made such an undertaking possible. During these first three weeks of the season, the bulk of the extractions focused on the objects from the new excavations in Pacheia Ammos at both the Industrial Area and the Rock Shelter. Kathy Hall, INSTAP senior conservator, and I instituted a pilot program for these extractions by studying stone tools for organic residues and phytoliths. With the help of Heidi Dierckx, we selected 23 stone objects from the Pacheia Ammos Rock Shelter (plus a quern from Priniatikos Pyrgos) as

likely candidates for producing organic residues and phytoliths. Overall, we garnered four samples from each object: mechanical scrapings from the exterior, solvent extractions for organic residue analysis, filtrate by-products from the solvent extractions, and particulates produced by acid washing the stone objects. In addition to the 23 organic residue samples extracted from the Rock Shelter for the pilot study, we extracted 344 samples from the Industrial Area, bringing the total number of samples in ARCHEM's Pacheia Ammos archive to 833.

With over 2500 samples in its collection, we were ready to develop the next step in the research protocol, the instrumentation process. We purchased a centrifugal evaporator with the express purpose of preparing 0.3ml subsamples for chromatographic analysis. Up to 36 vials at a time are simultaneously evaporated using the centrifugal evaporator and transported to labs overseas, where the residues are put back into solution upon arrival using dichloromethane before injection into a Gas Chromatography/Mass Spectrometry instrument. Presently, this research will take place at the Detroit Institute of Arts Department of Conservation Science thanks to a generous Mellon Foundation grant. We have made a tentative agreement with Dr. Christophe Vallianos, Director of the Museum of Cretan Ethnology's Research Center at Vori. In the near future, the ARCHEM director will help set up their GC/MS instrument, and for a modest price, have continued access to it.

This year, we also decided to branch out with additional analytical techniques. Demetrios Anglos, a senior researcher at the Foundation of Research and Technology,

Hellas (FORTH), obtained samples from ARCHEM that were analyzed by Nuclear Magnetic Resonance (NMR). We intend to continue collaboration of this nature in the near future. Samples will also be analyzed this year at the Detroit Institute of Arts using Raman spectroscopy, a nearly instantaneous method of qualitatively and quantitatively identifying compounds through the use of a laser. Though it will never produce the detail of GC/MS analysis, it is a promising technique that immediately identifies the general characteristics of organic residue samples in solution, as the instrument has the ability to identify compounds through transparent materials—in this case, our glass vials. Combined with our more detailed methodology and new techniques, ARCHEM remains a promising project that will continue into the future. Accordingly, our two major goals for next season are to continue refining the extraction procedure while also expanding the instrumentation and interpretation processes. Ultimately, these efforts promise to solidify ARCHEM's continued contribution to Aegean archaeology.

Total Tally of Samples for Summer 2007 by Site

Azoria	9
Mitrou	36
Mochlos	56
Pacheia Ammos	367
Palaikastro	8
Pyrniatikos Pyrgos	4
Trypetos	7
487 Total	

DISASTER PLANNING AND RECOVERY

By Eleanor Huffman, Assistant to the Director

In April, thanks to the recently initiated Personnel Development allowance and the generosity of Gerry Gesell, I traveled to London to participate in a disaster planning and recovery seminar, and now I am writing a plan for use in the event of a disaster in the building. The plan focuses on people first, then the antiquities storage, archives, and library. Everyone employed in the building will be involved at some level and the plan will be practiced with dry runs so people know what to do if necessary. As designated safety technician of the building, one of my goals is to avert disasters or accidents, but disasters such as those brought on by earthquakes or forces of nature must be managed after the fact. Good preparation will ensure that the best choices can be made immediately, and damage may be mitigated.

MESSAGE FROM THE DIRECTOR

It is a pleasure to highlight the successful work of the staff and members of the Study Center and our colleagues in the wider archaeological community working in east Crete in 2007. Several member projects, Azoria, Mochlos, and Pseira, conducted successful study seasons. The team preparing the pottery from the LM I houses of Block C at Mochlos identified two pottery fragments incised with Linear A, including a sign for olive oil. The Study Center also hosted or assisted several visiting teams studying pottery from several sites in eastern Crete—the Makrygiallos Project under the direction of Prof. E. Mantzourani from the University of Athens, and the Papadiokambos Project directed by Chrysa Sofianou of the 24th branch of the Greek Archaeological Service.

In July, the Study Center also hosted the new Irish team under the direction of Dr. B. Malloy excavating at Priniatikos Pyrgos with the support of Dr. B. Hayden of the University of Pennsylvania Museum. The Study Center's Chief Conservator, Dr. Stefi Chlouveraki, continues to lead the team conserving the pottery from the MM II shipwreck excavated near Pseira by Dr. E. Hadjidaki of the Underwater Antiquities Department of the Archaeological Service. I also want to mention the small rescue excavation of Minoan remains conducted by the 24th branch of the Greek Archaeological Service in fields several hundred meters north of the Study Center. We look forward to hearing more about these finds next year when the work is finished.

The summer lecture series included lectures by three Aegean Bronze Age Scholars. Prof. Jeffrey Soles presented a lecture on the Minoan reverence for their ancestors, and Dr. Melissa Eaby gave a paper on East Cretan Iron Age burial practices. Kostas Halikias also spoke about his Extra Urban Survey of Bronze Age Ierapetra.



Stephania Chlouveraki (right) and Matina Tzari (left) mend a unique EM pithos from Aphrodis's Kephali.

During the year the Study Center also hosted three workshops. Two were held in May at the Study Center, the first by Prof. S. O'Conner of Bradford University taught our conservation staff how to use the new Faxitron X-ray unit. The second workshop was by Prof. S. Manning, and focused on the collection of wood samples for analysis at the Malcolm and Carolyn Wiener Laboratory for Dendrochronology at

Cornell University. The third workshop was held at the Danish School in Athens in July. It brought together 30 scholars with unpublished LM IB pottery in an attempt to examine two critical issues: regional ceramic production and possible subphasing of the LM IB period. I want to thank my co-host, Erik Hallager, for making the event such a success.

There have also been some exciting developments to the physical plant in Pacheia Ammos. No change was more appreciated by our members than the appearance of high-speed internet service in June 2007. This system replaced the dial-up system brought to the village by the Study Center in fall of 1996; however, it is difficult to say which change was more dramatic—the high speed color



The extended 9 meter crane for aerial photography for the Photography Laboratory.

photos of Googleearth today or the black and white text reporting the news on PINE in 1996. The Photography and Conservation Laboratories also received two important upgrades. The first is a crane, ordered by Publication Team Photographer Chronis Papanikolopoulos, that allows us to lift a camera 9 meters above an excavated site. The second was a Faxitron cabinet X-ray machine discovered by Publication Team conservator Kathy Hall (see her report in this issue of Kentro).

The construction of a proper archive in the basement of the Study Center promises to provide an important stimulus for the preservation of our paper and electronic records. This work is described by Eleanor Huffmann elsewhere, but we all owe a great debt of gratitude to our library intern, Dr. Eletheria Daleziou, who oversaw both the construction of the space and organization and installation of the growing body of excavation and administrative records. The transfer of these records also allowed for addition of two large oak tables in the library, increasing the space for readers from 6 to 10. Finally, I want to report on the successful construction of the Kommos Apotheke

in Pitsidia. Several members of the Study Center helped the Director oversee its construction, and I would especially like to thank Publication Team Conservator Kathy Hall for overseeing the complicated transfer of antiquities to the new facility.

While the Study Center was busy with all of these activities, the highlight for many of us was a pair of parties in early June. The first was the wedding reception of Dr. Yiannis Papadatos and Dr. Eleni Nodarou, Director of our MacDonald Petrography Laboratory. We all wish them many years of happiness together. Our second party celebrated the 75th birthday of Prof. Gerry Gesell. The party for Gerry brought together her many friends and colleagues to see her receive a Festchrift devoted to Minoan religion.

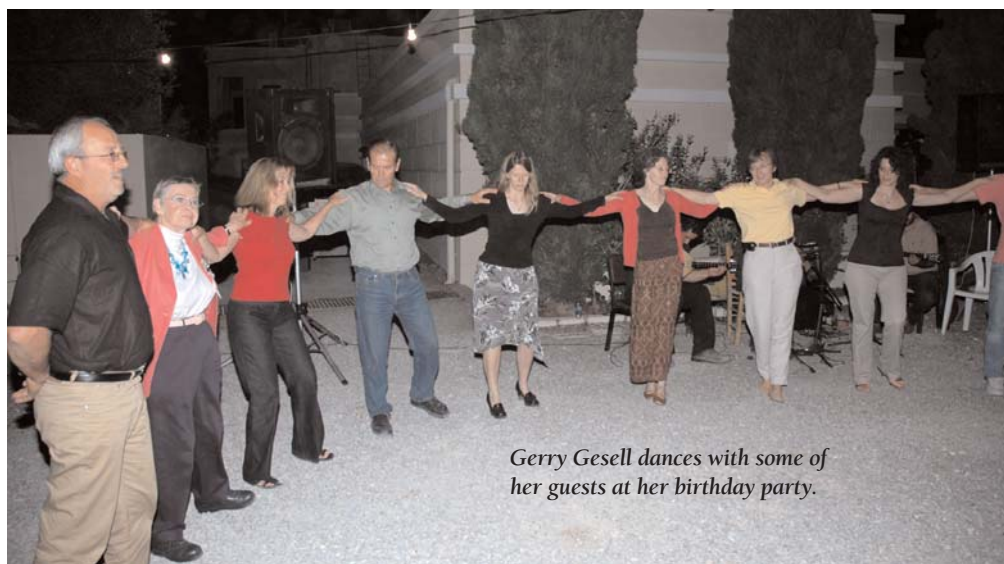
- Thomas Brogan



Sturt Manning shows Stephania Chlouveraki wood core sampling techniques.



Richard and Janette Sias, Gerry Gesell, Jeff Soles, and Metaxia Tsipopoulou at Gerry's 75th birthday party.



Gerry Gesell dances with some of her guests at her birthday party.



Yuki Furuya, the 2007–2008 INSTAP Librarian Fellow, in our newly re-designed library.



Joe and Maria Shaw, Tommy Brogan, and Tom Brogan outside the Kommos Apotheke.

THROWING SMALL VESSELS IN THE LM IB MOCHLOS POTTER'S PIT

By Jerolyn E. Morrison and Douglas P. Park

...man's first wheels used in the early cultures of Egypt, Greece and Mesopotamia... consisted of a disk, mounted on a shaft or a stone in the ground and were turned by means of a long stick or by hand. The potter could stand or stoop to the clay wobbling in the centre of the disk while he worked alternately at trueing up the mound revolving the wheel.

--Peterson describing Shoji Hamada's hand wheel, 1974: 233

This summer we gave a spin at figuring out Minoan wheel technology by turning our energies to the potter's pit in the Mochlos Artisan's Quar-

ters . Being both curious students of anthropological archaeology and creative potters, we are interested in developing experimental programs to recognize ceramic technologies and shifts in ceramic production in order to discuss cultural change in the archaeological record. To begin our study, we looked at

the two primary needs of the craftsperson, a suitable workspace and a proper tool kit. To understand these two primary needs, we posed the following two questions: was the exterior semicircular space behind Building A in the Artisan's Quarters used as a potter's pit, and can the centrifugal force created by Evelyn's Type 3C 'flywheel' (1988, 83-126) maintain enough energy to produce the pottery characteristic of the Bronze Age?

Archaeological Context of Minoan Craft Activity

Soles has identified a LM IB Minoan stone lined potter's pit located in the Mochlos Artisan's Quarters (Soles

2003: 37). The potter's pit is a semicircular exterior space that is roughly two meters wide by three-quarters of a meter deep and attached to the south side of Room 4 and along the west side of Room 10 of Building A. The presence of a wooden support beam is hypothesized. Two sizeable notches set

within the wall architecture of the potter's pit would have allowed a support beam to lie securely across the upper course of stone, thus supporting the wheel axle in a vertical position for potting (Figure 1). Other finds associated with potting activity unearthed in the pit are one ceramic bat, two polishers used for shaping the vessel, one pivot stone to support the wheel axle, one obsidian blade and core used for cutting string, used by the potter to remove the newly potted vessel from the bat or the wheel (Soles 2003: 36-38).

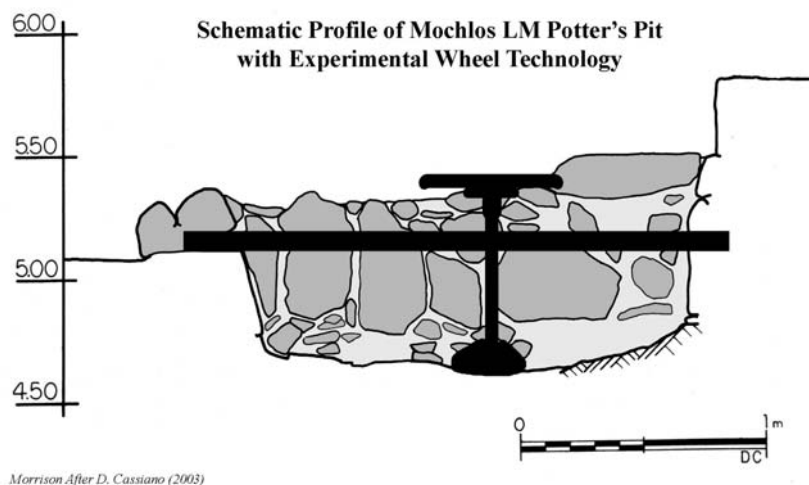


Figure 1

The potters working in the Artisan's Quarters used two different types of large, heavy, flat-topped wheels, Evelyn's Types 3A and 3C (Evelyn 1988: 100–106). We believe that the existence of the two wheel types at a single-phase site, such as the Mochlos Artisan's Quarters, indicates that the potters used a variety of wheel technologies to perform different tasks. We are currently developing experimental tests for Mochlos wheel technology that will explore these technologies in pottery production. The undersides of the wheel types are the distinguishing characteristic that indicates the Minoans had developed different wheel technologies. The underside of Type 3A is constructed so that the rim and wheel are an even thickness with a very shallow central socket. The underside of Type 3C is constructed with a more elaborate thickened 'flywheel' rim with an attached deep scored socket. Clearly these wheels required different mechanism for installation and use.

Experimental Procedure and Results

Many scholars have commented on the installation and use of the Type 3C wheel and believe it can be turned to maintain a speed that would form a pot. With theories of the mechanical installation of the wheel in mind and a replica copy of a Type 3C wheel and Type 4 collar (Evelyn 1988: 107), compliments of Don Evelyn, we headed to the Artisan's Quarters to rebuild the Mochlos potter's pit and learn how to operate the wheel.

Based on architectural features found in the semi-circular pit behind Building A (Soles 2003), we designed a wooden support beam that could hold the weight and withstand the turning force of the potter's wheel (Figure 1). Once we were confident of the design, we installed it in the pit and secured it with wet clay. Below the wooden support beam, we placed our replica of the pivot stone in the same location it was found in the pit. With our architectural support complete, we began the installation of the experimental wheel. We first inserted the rounded tip of the wooden axle into the socket of the wheel. Then we secured the collar to the wheel by adding clay in and around the deep socket and threading the collar through the axle until it made contact with the underside of the wheel (See Evelyn 1988: 109, Figure 10). Once the wheel apparatus was secure, we placed it through a padded hole in the support beam so that the pointed end of the axle rested in the shallow "V" groove



Figure 2



Figure 3

of the pivot stone. We then leveled the wheel-head by aligning the wooden axle and pivot stone so that it could rotate freely (Figure 2).

In an attempt to develop the skills to make a pot on a hand-turned wheel while standing upright, we consulted a variety of video clips and photographs taken of potters using hand-turned wheels. However, in the end we found Japanese master potter Shoji Hamada's writings the most useful. In his writings he emphasizes the importance of finding the balance between your hands, the clay and the wheel. To find the balance, you must relearn how to position your body while setting the tempo and rhythm between releasing the clay and spinning the wheel (Figures 3 and 4). The process is both physically and mentally frustrating, but in the end extremely satisfactory (Figure 7).

The basic procedure we used to throw on the Minoan wheel is fairly simple (see sequential steps in Figure 6). The potter first lubricates the wooden axle with olive oil at the point where it comes in contact with the wooden support beam and pivot stone. Other lubrication materials could have been used, such as other types of oil or crushed graphite. We choose olive oil since it was the most available. Next, the potter wets the flat top surface of the wheel head and adds the wedged clay at its center. With feet a shoulder's width apart and knees slightly bent, the potter places his right hand across his/her body to grasp the rim of the wheel-head to spin it as fast as it will turn. The potter then centers the clay on the wheel-head by firmly holding the clay wedge with the left hand while spinning the wheel with the right hand. Once the clay is centered, the potter alternates between spinning the wheel with his/her right hand and throwing the vessel using both hands. When the form of the pot is complete, the potter uses a thin piece of string to separate the base of pot with the flat



Figure 4

top of the wheel head. The pot is then removed and set aside for decoration and drying before it is fired and becomes a ceramic vessel.

Conclusion

We wanted to know how Mochlos potters produced small vessels, such as cups and bowls, in the Artisan's Quarters using a wheel that belonged to Evelyn's Type 3C class. To do this, we developed an experimental test that would allow us to understand how the Mochlos potters could have designed their workspace and used their tool kit. The following three discoveries were made: (1) the semicircular pit behind Building A could

Figure 5



perfectly accommodate Minoan-sized people to make pottery, (2) we were able to make vessels that resembled conical and ogival cups within the first week of operating the wheel, and (3) it was easier to produce a vessel on the wheel with one person operating it rather than two.

We discovered that the semicircular pit behind Building A is perfectly designed to accommodate the production of pottery using a hand wheel. The exterior location of the potter's pit offers plenty of natural light so that one can see the forms being potted; the south wall of Room 4 and the west wall of Room 10 would have provided ample protection from the sun and the strong sea breezes. Ergonomically, the pit is only

designed for Minoan-sized people, as both of us are no taller than the average height of a Minoan individual. The flat top of the wheel hits us both exactly at our center of gravity, which is at our waist. It is essential to use one's center of gravity to find and maintain the proper rhythmic balance between the speed of the wheel and the pressure one puts on the clay when potting. We made this discovery when we watched a taller, well-trained potter working on the experimental wheel. He had trouble centering and producing forms on the wheel since he was bent over and could not maintain the proper balance.

We spent the shaded afternoons, from around 4:00 to 8:00 on the last 2 weeks in July working in the potter's pit producing vessels. This is not a concentrated amount of time, yet we were still able to produce vessels that resembled Minoan cups. During the first afternoon in the potter's pit, we installed the wheel and began working. It did not take very long to realize that one must find the correct balance between the positions of our bodies, spinning the wheel and forming the vessel. Finding this centered balance

was the most difficult task of the experiment. By the end of the first afternoon we had begun making crude conical cups and shallow bowls. By the end of the first week we succeeded in making shapes that resembled ogival cups (Figure 7).

We designed and installed our wheel to be turned by spinning the rim of the wheel-head. We did not install a turning cross-bar, like the Thrapsano pithos wheel, since it is a well known technology. Also, another potter in Evely and Morrison's experimental program is learning how to make adjustments on this sort for potting small vessels. The wheel-head is large enough that two people could operate the wheel. The master potter would stand on one side of the wheel to



Figure 6

form the vessel while the other person would stand opposite the potter and spin the wheel-head by turning the rim (Figure 5). As noted above, we found that operating the wheel in this manner was very awkward. It was much easier for one person to keep the balance between spinning and potting. We concluded that if the Minoan potter operated the wheel by spinning the rim, he or she could easily produce a pot without the aid of an assistant.

Experimental Archaeology and Anthropology in Bronze Age Studies

The most practical way one can understand ancient technology is to build it, use it, experience the inherent setbacks, and make informed adjustments. Through experimentation we reconstructed how a Minoan potter at Mochlos would have produced cups, bowls, small jars, and jugs. Understanding how potters organized themselves to produce pottery is one insight into how Minoan craftsmen and craftswomen thought about their environment and their trade. Experimental archaeology and anthropological theory in Bronze Age studies are central to the reconstruction of all aspects of human activity. The study of the cultural process of human activity is a field of research that has the ability to breathe life into the archaeological record, for it brings the dynamic to the static.



Figure 7

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THE STUDY CENTER ARCHIVE

By Eleanor Huffman

A new record's management era began for the INSTAP Study Center with the completion of the Archive, which now houses the paper and digital records, site plans, and artifact drawings of the excavation projects served by the Center, the Center's business records, and some library holdings. These records are now housed in a safe, humidity-controlled and pest-free environment. The Archive room was constructed in the basement in an area formerly occupied by three storage cages used by projects to store equipment. After the physical construction, the room was furnished with filing cabinets, map drawers, a dehumidifier, and a work table. Sturdy, freshly painted blue shelves were attached around the walls. A local metal artisan built a custom-designed cart for use when retrieving and replacing records. In conjunction with the construction of the archive, Eleftheria Daleziou, the 2006–2007 Library Fellow, helped us reorganize our record storage system and work out a record retention policy.



Eleftheria Daleziou in the Archive at the beginning of the project.



Eleanor Huffman at work in the completed Archive.

THE NEW X-RAY UNIT AT THE STUDY CENTER

By Kathy Hall

This year we have been testing the new Faxitron X-ray unit in the Conservation Laboratory (Figure 1). The unit offers exciting opportunities for technological studies of many different materials as well as helping conservators in the preservation of artifacts.

X-radiography is commonly used for the study of many material types including metals (excepting lead), ceramic, glass, textile, seeds/other archaeobotanical material, and of course, human and animal bone. There are also geoarchaeological applications with soils and sediments (for example, the identification of hammerscale from smithing). X-radiography has also been used in archaeology in the examination of soil blocks and sealed vessels which may contain complex groups of artifacts.

Among the most common uses of X-radiography of cultural materials are:

- identification of an object and its condition.
- identification of manufacturing method
- identification of finishing methods and decoration
- identification of faults, breaks, repairs and reuse

X-radiography is a non-destructive, photographic technique in which high energy radiation is generated and directed at an artifact placed on a piece of radiographic film. Since regions of different densities absorb differing amounts of X-rays, the developed plate gives an image of the internal structure of the artifact.

The INSTAP Study Center's unit is an industrial machine with a range of 20–110 kVp which produces low intensity X-rays useful for the study of bone and ceramic material as well as higher intensity X-rays suitable for the study of metallic artifacts. Timing of exposure is also very flexible. The unit produces X-rays in a totally enclosed, radiation-proof cabinet which is checked for radiation leaks annually by a qualified medical physicist.



Figure 1: The Faxitron X-ray unit

Examples of use at INSTAP

Michel Roggenbucke has been working with Gerry Gesell in her technological examination of the goddess figurines from Vronta. X-radiography has proved to be a valuable help in understanding how the figurines were put together (Figures 2 and 3). The radiograph shows as extra wedge of clay added to strengthen the thumb, as well as the method of attachment of the fingers through insertion into the palm. The direction of the voids in the ceramic fingers shows how the clay was rolled. Dowels added by the conservator to strengthen joints are also visible.

The unit has also been used for the study of a ceramic assemblage by Dr. Ina Berg at Knossos (forthcoming) and a study of selected weapons from the Knossos North Ceme-

tery. An investigation of all metallic artifacts from the site of Vronda is in progress.

X-radiography is available to all researchers at INSTAP. To schedule investigative X-radiography of archaeological material please contact Dr. Tom Brogan. The process produces pieces of X-ray film size 18 x 24 cm or 24 x 30 cm. If required for publication, images can be photographed on a light table by the INSTAP photographer, Chronis Papanikolopoulos.

After purchase of the unit, a short professional development course was organized at the INSTAP Study Center. "Cultural Material X-Radiography; Imaging Techniques, Interpretation and Digitisation" was taught on the 12th and 13th of May by Sonia O'Connor of the Department of Archaeological Sciences at the University of Bradford. Attendees included INSTAP conservators Michel Roggenbucke, Steffi Chlouveraki, and myself, as well as Kleio Zervaki (Haghios Nikolaos Museum) and Karen Loven (Agora Excavations).

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Faxitron unit/osteological examples:

http://www.faxitron.com/43855a_gs.htm

Metal artifacts: http://www.english-heritage.org.uk/upload/pdf/X_Radiography.pdf



Figure 2: The hand of a goddess figurine from Vronda seen in regular light (photograph by Kathy Hall)

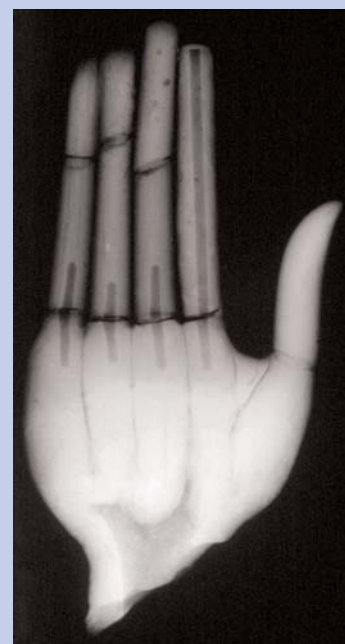


Figure 3: A radiograph by Michel Roggenbucke of the hand (photograph of the radiograph by Chronis Papanikolopoulos)

EXPERIMENTS WITH MINOAN INCENSE BURNERS

By Pete and Gracia Travis

Like many interesting lines of research, it all happened by accident. We visited Crete for the first time in the spring of 2006, partly to see our friends John Lewis and Rita Roberts, but also to track down and photograph the Rock Rose of Crete in its natural habitat. While it may seem strange that we would fly from our home in England to look at one fairly common flower, the Rock Rose produces Labdanum, an aromatic substance sometimes used in incense and perfume, and we trade in incense resins. Naturally enough, we visited the Hagios Nikolaos museum, and there we observed more incense burners than we had ever seen in a single exhibition. Some of them, however, were very odd; Rita and John (who work as volunteers at the INSTAP Study Center for East Crete) they suggested that we should bring this to the attention of Thomas Brogan, the Center's Director. We did this by scrawling rough designs on a bit of scrap paper and presenting them to him. In response to our drawings, he suggested that when we returned to the UK, we could have some replica burners made, and then simply try them out. He also recommended that we should liaise with archaeochemist, Andrew Koh, and put together an experimental archaeology project. We have a colleague, Jim Newbould, who specializes in making replicas of historic (and prehistoric) pottery, and he produced replicas of three designs using both a hand turned slow wheel to shape a clay type matched to the original artifacts (Figure 1). We returned to Crete a few weeks later to meet Andrew, bringing with us samples of some resins that we believed would have been available in Bronze Age Crete. We then introduced Tom, Andrew, and several other team members to the amazing aromas of natural incense.



Figure 1: Jim Newbould makes a replica of a Minoan incense burner.



Figure 2: Our replica of the Type A incense burner.

The Burners

Firstly, the Type A burner (Figure 2) is a rather curious shape, consisting of a saucer with a dome in the middle. Sometimes this dome is perforated. Our replica

was based on Late Minoan IB artifacts from the Hagios Nikolaos Museum. The obvious method of usage was to put lighted charcoal into the cavity, but this turned out to be a total failure, as not enough heat could be generated to vaporize anything, even with the use of bellows! When the bowl was filled to the rim with burning charcoal, however, the charcoal remained hot enough to vaporize *B. frerana* resin for about two hours (Figure 3). Further air flow increased the temperature of the coals considerably, and as a result windy conditions instantly ignited the resin. We found that there was no need to use bellows, as the perforated inner dome appeared to amplify the air-flow to the charcoal. Even so, we did notice that the base of the burner became too hot to hold, and managed, in windy conditions, to scorch the wooden surface on which it was resting. Accordingly, we decided that we needed a way to carry the lit burner. We soon found that a small basket of willow proved to be the ideal solution. We filled a burner with hot charcoal and placed it in the basket (Figure 4). We soon noticed that we could hold the basket quite comfortably, and when indoors, the willow was only slightly scorched. Interestingly, this combination of basket and burner resembles the incense burner held by the “Young Priestess” in a wall painting in the West House at Akrotiri, Thera (Figure 5).

The Type B burner (Figure 6) bears a close resemblance to a modern burner, and as a result, it was only natural for us to conclude that it was



Figure 3: Our replica of a Type A incense burner filled with burning charcoal.



Figure 4: The replica of a Type A incense burner filled with burning charcoal is placed in a basket.



Figure 5: A detail of the “Young Priestess” fresco from the West House at Akrotiri, Thera (from Christos Doumas’ *The Wall Paintings of Thera*, Athens, 1992).

an incense burner. Our replica is based both on a Late Minoan III artifact from Sitia and on the illustrations in Georgio’s 1979 article “Late Minoan Incense Burners.” In all, five were made, but they were left undecorated in order to avoid unnecessary contamination. To test the effectiveness of the burners, the replicas were filled with ordinary lumpwood charcoal purchased at the supermarket. We then added an incense resin. The results were amazing, as the charcoal in the burners retained enough heat to burn for up to six hours unattended. Apparently, the perforated lid slows down the airflow to the charcoal. Although the base certainly became very hot, the unit was very easily moved around by the handle. This characteristic of the burner led us to believe that it could have been used for another purpose. We hypothesized that, given the precious and sacred nature of fire in the Bronze Age, these receptacles could have been designed to retain fire while it was carried from room to room or from house to house. If used for this purpose, the lid would have acted as a guard or “cur-few” (literally, ‘cover fire’), thus preventing materials from falling into the vessel and catching fire. This question could be answered by residue analysis of deposits on the inside of the lids.

We did our initial tests on a single burner. In order to replicate residue deposits, we employed the remaining four burners. Each was filled with charcoal, and over a period of two

hours, a different type of incense was burned in each burner (Figure 7).

The main two resins used for all the experiments were types which would have been readily available to the Minoans: Labdanum (*Cistus* species), which grows on Crete, and Mastic (*Pistacia lentiscus*), a plant which now comes from Chios but is known to have grown on Santorini (Friedrich 1978, 109–128). Both vaporize at a relatively low temperature. Mastic is similar, if not identical, to the “terebinth” found on the Uluburun ship wreck and identified at Tel Armana in Egypt (Serpico and White 2000a and b). Furthermore, Andrew Koh has already discovered traces of labdanum on some Minoan artifacts.

The other two incenses we used for this experiment were Black Storax and Coptic Frankincense. Black storax is a mixture of the charcoaled bark and resin from *Liquidambar orientalis* which comes from the Levant (nowadays mainly modern Turkey). The potential use of Coptic Frankincense (*Boswellia frerana*) during the Bronze Age is debatable. Although it comes from Somalia, we believe that the potential trade routes to the eastern coast of Africa would have been easier than those needed to obtain the Arabian Frankincense (*B.sacra*)



Figure 6: Our replica of the Type B incense burner.

Our replica Type C burner (Figure 8) was based on a Late Minoan IB artifact from the Makryialos Minoan villa (the same provenance as Type A). This burner was the item that had originally caught our attention, and as we initially predicted, burning charcoal in the lower cavity failed to achieve anything more than a melted resin (Figure 9). Even so, one extraordinary result was obtained. If one fills the top bowl with burning charcoal, enough heat radiates downwards to release the aromas from resins placed in the lower cavity (Figure

10). However, once again, residue analysis would be needed to give us a better understanding of how this item was employed.

Conclusion

Bringing these ancient artifacts back to life was a fascinating challenge. Hopefully these experiments will help to build a working hypothesis on the functionality of Minoan incense burners. At this point, all the replicas have now been given to the Study Center for further anal-



Figure 7: Testing the Type B burners.

ysis and comparison. If we have used these replicas in the correct manner, then our own burners should possess residue deposits which would correspond with those on the original artifacts. But if this does not occur, then we will just have to try something else!

Acknowledgements

We would like to thank all the staff at the INSTAP Study Center for their hospitality and help. We would like to extend a special thanks to Tom Brogan and Andrew Koh for their boundless enthusiasm about what must have seemed at times to be a very eccentric project. We very much hope that we will have the pleasure of being able to continue working with everyone in the future.

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Figure 8: Our replica of a Type C incense burner.



Figure 9: Burning charcoal is placed in the lower cavity of the replica of a Type C incense burner.

Pete and Gracia Travis are freelance incense specialists who work with the UK Heritage Industry.

Their website is www.panspantry.co.uk





Figure 10: The top bowl of the Type C replica is filled with burning charcoal.

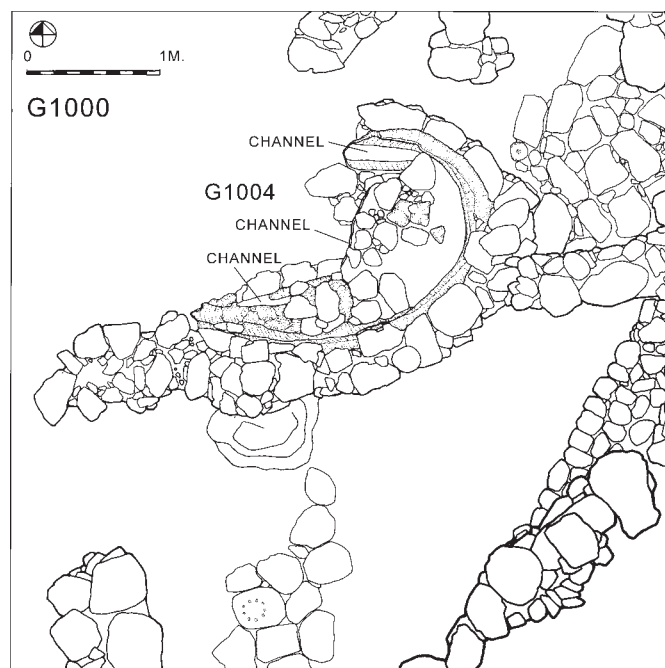
THE PRINIATIKOS PYRGOS PROJECT, 2006–2007

By Barbara Hayden, University of Pennsylvania Museum

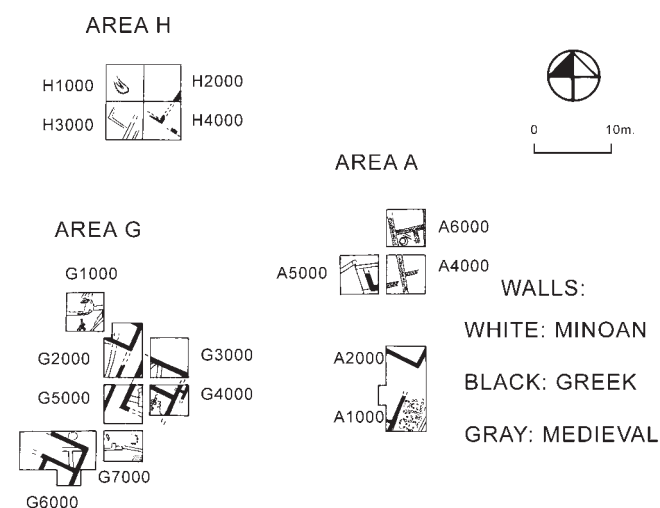
One of the goals of the Vrokastro Survey, which I conducted with Jennifer Moody (Baylor University) was to select a site whose excavation would develop and refine the conclusions established through this survey concerning the cultural development of the region. (These conclusions were published in *The Vrokastro Reports* 2, 3). The criteria involved a well-preserved settlement with substantive prehistoric and historical phases, evidence for industrial/economic activities, and trade networks. The harbor settlement of Priniatikos Pyrgos, on a sandy limestone headland at the northern end of the Istron River Valley, was chosen for further investigation, including excavation. This selection was based on its extensive chronological range, size, evidence for ceramic production, and complexity and quality of artifacts recovered through survey.

A brief rescue excavation conducted in 2005 by Dr. Metaxia Tsipopoulou (Senior Archaeologist, 24th Ephorate) with Hayden uncovered two ceramic kilns along the western side of the headland. The larger kiln is cross-draft channel type, and its form and associated pottery suggest its date as early Neopalatial (Figure 1); EM IIB-MM IA ceramics found within the smaller kiln on the northwest slopes, and on and below a nearby soil and pebble surface (in trench H3000), suggest the possibility of an earlier date for this smaller kiln.

Remote sensing undertaken in the Istron coastal zone from 2002–2005 by Dr. Apostolos Sarris (Institute for Mediterranean Studies, Rethymnon) also



Plan of cross-draft, three-channel kiln in trench G1000. The western half of the kiln has been eroded by the sea. Clay and cobble surface preserved at the eastern back of the kiln preserves a surface extending above the channels.



Plan showing the trenches and walls revealed in Areas G and A, from the western edge of the promontory to its center.

indicated that Priniatikos Pyrgos was part of an overall coastal settlement extending east to include the Greek harbor settlement of Istron, on the large promontory of Nisi Pandeimon (identified through survey). The well-preserved Neopalatial (and Final Neolithic/Early Minoan I) part of this settlement was also identified by Sarris in the Kambos, located under a modern football field. The chronology of the architecture revealed through remote sensing was established from sherds brought to the surface by trenching and coring, undertaken as part of a study of the geology and geomorphology of the coastal zone undertaken by Dr. Iannis Bassiakos, Konstantinos Athanassas (Demokritos University), and Dr. Kosmas Pavlopoulos (Harokopio University).

The rescue excavation continued in 2006 in a series of trenches excavated within the G area, along the western side of the promontory, and in the A area, in the central part of the headland (Figure 2). Near the surface in the G area, a Greek architectural complex, contemporary with the Greek harbor settlement on Nisi Pandeimon, was uncovered. The design of the Greek settlement involved construction of long, parallel walls, which in one case frame a road through the settlement. Houses and rooms are built against these long walls (as in trenches G3000, G4000, and G5000). The courtyards and



Large Greek drinking cup, cat. no. 07-0662.



Sherd from a large vessel found in trench H4000, showing part of a male face in profile, 7th c. B.C.

roads identified in this settlement were paved, possibly in the Early Hellenistic period. At least two or three major phases of Greek settlement have been identified through ceramic studies conducted by Dr. Brice Erickson, one dated to 475–450 B.C., and identified by ceramics found in a large ash deposit within trench G2000 and along the north side of trench G5000. An early Hellenistic phase is seen in pottery recovered from the large Greek structures revealed in trenches G3000–G6000; this phase was also identified in a Greek midden found directly north of the Greek room identified in trench G6000, sealed beneath a paved court. These phases are also seen in a pottery dump recovered from an open area south of a Greek building in the central part of the site (trench A2000), and in the corner of a Greek building found in trench A5000. A Cretan high-necked drinking cup (Figure 3), unique for its large size, was recovered from the dump in trench A2000. Although the Classical and Early Hellenistic

periods represent major phases of occupation, ceramics from other periods, including the Early Iron Age, Orientalizing-Archaic (Figure 4), and the Late Hellenistic period (seen in Megarian bowl fragments) are represented at the site. This Late Hellenistic phase is also seen in sherds from a Greek building in trench A2000.

A deep and complex Bronze Age sequence was also excavated within area G in 2006. These phases begin on bedrock with a stratum of Final Neolithic and Early Minoan I pottery found with burned soil,

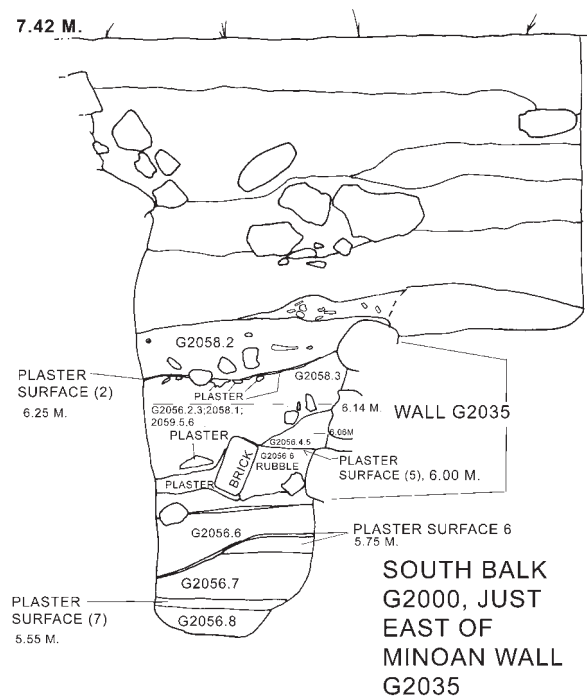
large pieces of carbonized wood, bone, and large obsidian blades. Within the western part of trench G2000, for example, an Early Minoan wall (G2035) evidencing at least two construction phases is associated with as many as four strata. These contain or are defined by plaster surfaces, two of which occur below wall G2035, one in the southern balk of the trench, and one directly below the wall (Figure 5a,b). Occupation of the site in Middle Minoan I-II is evidenced above these deep Early Minoan strata throughout the G area in strata; Middle Bronze Age sherds are sometimes associated with prepared plaster or pebble surfaces. The elevation of Late Minoan strata indicate that in many places, Greek rebuilding must have destroyed construction belonging to this Neopalatial phase, but the abundant ceramics of this period suggest that the settlement belonged primarily to the early part of this period—although some Post-palatial ceramics have been found, especially in trenches G1000 and G4000.

Several rhyton fragments (Figure 6) found in trenches G4000 and A4000 suggest that special-function areas occurred within the Neopalatial settlement. Trench G7000, however, may have been an open area within the Greek settlement, and a sequence of Late Bronze Age surfaces have been identified within this trench, located very close to the modern surface (Figure 7). This trench can therefore furnish important data concerning the development of the Neopalatial settlement, since Late Minoan surfaces and strata have not been disturbed. This chronological sequence is also attested in the A area, with Early and possibly Late Minoan walls recovered in trenches A1000 and A2000, and Early Minoan pebble strata—perhaps surfaces—recovered in trench A5000 (Figure 8). Although the evidence for early Neopalatial occupation on Priniatikos Pyrgos is extensive, this area must represent only a part of the overall settlement, which has also been observed in the well-preserved walls and pottery revealed through remote sensing and trenching in the Kambos.

During a brief excavation season in 2007, a large, multi-phased building complex of probable Greek date



Base of wall EM G2035 showing EM plaster surface 6.



Section of south balk immediately east of wall G2035, showing deep EM plaster surface at 5.55 m. Surface 5 (6.00 m.): EM IIB; Surface 6 (5.75 m.): EM I-IIA; Surface 7 (5.55 m.): EM I-IIA (some FN).

(Figure 9) was revealed in trench II, just north of the A area. Even though these walls are located near the modern surface, segments of related, intact plaster floors are beginning to be found. One later burial (possibly Late Roman), in an area outlined by cobbles, has been found placed within this complex, and it is probable more burials will be identified. Even though this building complex is on the crest of the hill, overlying a higher bedrock profile, it is quite possible that the Greek walls are located over Bronze Age strata that may include fragmentary walls and surfaces. This stratigraphic sequence has been observed nearby, in the A area.

Because so many periods are represented in this part of the overall harbor settlement, a more complete historical sequence, involving both growth and decline, can be achieved. This long perspective involves the chronology and size of the initial settlement of the coast, the relative strength of the Early Minoan periods that follow, their developing ceramic traditions, and, through imports, the degree to which this site was linked to Mirabello, eastern Crete, and the Aegean. Initial study of the ceramics has already indicated a foundation date during EM I (if not earlier), and that a large, thriving EM IIB community continues into EM III and MM IA, and encompasses later periods of the Middle Bronze Age, as well as the early Neopalatial period. Analysis of Greek ceramics has refined the chronology of the earliest Greek settlement of the coast as established through survey. Pottery suggests that the polis of Istron may have been founded in the late 6th to the mid-5th century, or witnessed a significant growth during this period. The environmental framework



LM IA rhyton from trench G4000, cat. no. 07–0809.



Three high MM-LM IA pebble surfaces in the southeast corner of G7000 which continue in the eastern and southern balks; flat slab in eastern balk near the modern surface may indicate Greek surface.

this diachronic settlement is also being reconstructed through the Istron Geoarchaeological Project, alongside bone and botanical studies. For example, the presence of cattle bone associated with the overall Greek settlement relates to use of

the river delta, and possibly to the nature and extent of cultivation in the immediate environment of the Greek site.

This new data involves questions concerning the inception, duration, function, and economical development of settlement within the coastal zone and throughout the region. These questions, in turn, are derived from the results of the Vrokastro Survey. In this manner, new excavation completes the twenty years of fieldwork previously undertaken within this portion of Mirabello and serves as a platform for future exploration of this large, multi-faceted coastal emporium.

Acknowledgments

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EM I-IIA pebble surface (A5022) found in trench A5000, with a mud brick located on this surface.



2007 excavation of a Greek building complex on the crest of the promontory, from the northwest. Shown are two large rectangular rooms, roughly oriented east-west, with a large door leading into the southern room.

Trinity College and University College Dublin (2007), under the direction of Dr. Barry Molloy, excavated at the site and assisted with the laboratory work at the INSTAP Study Center. Architectural drawings have been provided by Damon Cassiano (2005) and Joe Griffin (2006, 2007), who also assisted in the stratigraphic study.

thank you!

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*Chronis Papanikolopoulos, Eleanor Huffman, and Evi Sikla
relax on the back porch of the Study Center.*

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